EXPERIMENT - 3

Performing MITM Attack Using Ettercap

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux virtual machine.
Login to the system.

Step 2: Check Network

Open terminal and verify IP address:

ifconfig

Ensure the attacker and target systems are in the same network.

Step 3: Launch Tool

Start Ettercap in graphical mode:

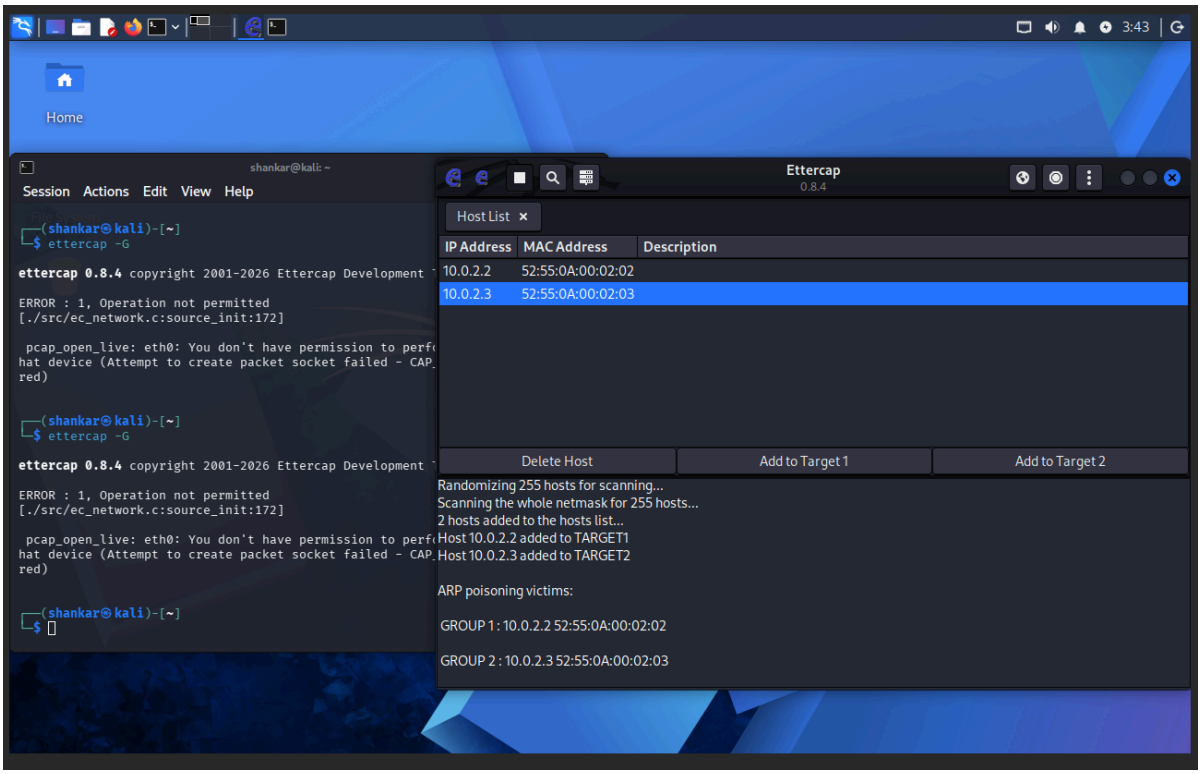
ettercap -G

Step 4: Scan and Select Targets

- Choose **Sniff → Unified Sniffing** and select interface
- Scan for hosts
- Identify:
    - Target system
    - Gateway (router)

Step 5: Perform MITM and Monitor

- Configure MITM using ARP poisoning (conceptually)
- Start sniffing
- Observe intercepted network traffic in the tool



Result

The MITM attack was successfully demonstrated in a controlled environment, showing how unsecured communication can be intercepted and why encryption is important.